1. Assume a 200mm diameter wafer has a cost of 12, contains 84 dies, and has 0.045
defects/cm2. Assume a 140mm diameter wafer has a cost of 15, contains 90 dies, and has 0.010 defects/cm2.
(a) Find the yield for both wafers (20%)
(b) For the 200mm diameter wafer, if the number of dies per wafer increased by 15% and the defect per area unit increases by 15%. Find the die area and yield. (10%)

$$\text{Yield} = \frac{1}{\left(1+\left(\text{Defects per cm}^2 \times \frac{\text{Die Area (cm}^2\text{)}}{2}\right)\right)^2}$$

(a)

200 mm : Die area : $\frac{10^2\pi}{84} = 3.74\ \text{cm}^2$ #

Yield : $\frac{1}{\left(1+\left(0.045 \times \frac{3.74}{2}\right)\right)^2} = 0.85$ #

140 mm : Die area : $\frac{7^2\pi}{90} = 1.71\ \text{cm}^2$ #

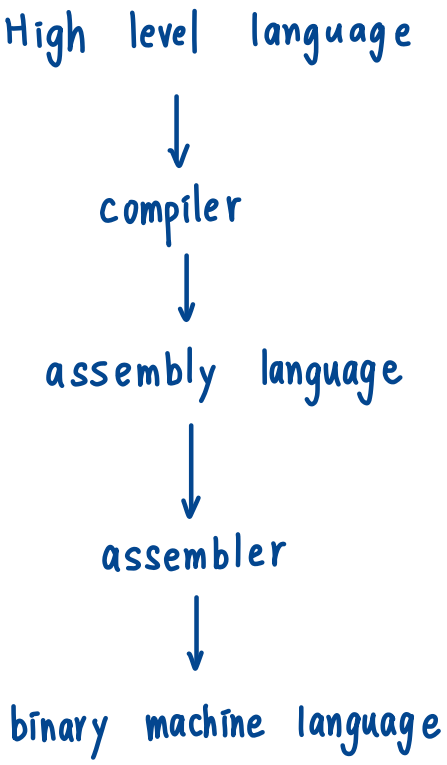
Yield : $\frac{1}{\left(1+\left(0.01 \times \frac{1.71}{2}\right)\right)^2} = 0.98$ #

(b)

200 mm : Die area : $\frac{10^2\pi}{84 \times 1.15} = 3.25$ #

Yield : $\frac{1}{\left(1+\left(0.045 \times 1.15 \times \frac{3.25}{2}\right)\right)^2} = 0.85$ #

2. Describe the steps that transform a program written in a high-level language such as C into a representation that is directly executed by a computer processor. (10%)



3. Consider two different implementations of the same instruction set architecture.
The instructions can be divided into five classes according to their CPI (class A, B, C, D), P1 with a clock rate of 2.5GHz and CPIs of 1,2,3,3 and P2 with a clock rate of 3GHz and CPI of 2,2,2,2. Given a program with a dynamic instruction count of 1.0e8 instructions divided into classes as follows: 10% class A, 20% class B, 50% class C, 20% class D
(a) What is the ratio of CPI for P1/P2? (10%)
(b) The result of the benchmark running on the machine has an instruction count of 2.5E12, and execution time of 900s, and a reference time of 9000s. Find the CPI if the clock cycle time is 0.25ns.(10%)
(c) Find the increase in CPU time if number of instructions of the benchmark is increased by 20% and the CPI is increased by 5%(15%)

	A	B	C	D	clock rate
P1	1	2	3	3	2.5 G
P2	2	2	2	2	3 G

(a) $\dfrac{P1:(1\times0.1+2\times0.2+3\times0.5+3\times0.2)}{P2:(2\times0.1+2\times0.2+2\times0.5+2\times0.2)} = \dfrac{2.6}{2} = 1.3$ #

(b) $\dfrac{900/0.25\times10^{9}}{2.5\times10^{12}} = \dfrac{3600\times10^{9}}{2.5\times10^{12}} = \dfrac{3.6}{2.5} = 1.44$ #

(c) CPU time = instruction count x CPI ÷ clock rate

⇒ $1.2\times1.05\div1 = 1.26$, increase 26% #

4. Assume for arithmetic, load/store and branch instructions, a processor has CPIs of 3, 12, and 5, respectively. Also assume that on a single processor a program requires the execution of 2.56x109 arithmetic instructions, 1.28x109 load/store instructions, and 256 million branch instructions.
(a) By how much must we improve the CPI of load/store instructions if we want the program to run two times faster? (10%)
(b) By how much is the execution time of the program improved if the CPI of arithmetic and branch instructions is reduced by 50% and the CPI of load/store is reduced by 20%? (15%)

(a)

$$\frac{2}{1} = \frac{(3 \times 2.56 + 12 \times 1.28 + 5 \times 0.256) \times 10^9}{(3 \times 2.56 + X \times 1.28 + 5 \times 0.256) \times 10^9}$$

$X = 2.5$ $\quad \frac{2.5}{12} = 0.21$, reduce 79% #

improve to CPI = 2.5 #

improve 79% #

(b)

Origin : $3 \times 2.56 \times 10^9 + 12 \times 1.28 \times 10^9 + 5 \times 0.256 \times 10^9 = 24.32 \times 10^9$

Improved : $(3 \times 0.5 \times 2.56 + 5 \times 0.5 \times 0.256 + 12 \times 0.8 \times 1.28) \times 10^9 = 16.77 \times 10^9$

$$\frac{16.77 \times 10^9}{24.32 \times 10^9} = 0.69$$, reduce 31 % #

1.45 times faster #

improve 31 % #